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Organic Electronics

Precisely Patterned Channels in a Vertical Organic Electrochemical Transistor with a Diazirine Photo-Crosslinker

Yueping Lai, Jingliang Cheng, Miao Xie, Jianhua Chen, Guichuan Zhu, Wei Huang, and Liang-Wen Feng**

Abstract: Organic electrochemical transistors (OECTs) rely on both efficient ionic doping/de-doping process and carrier transport in the mixed ionic-electronic channel under the modulation of gate bias. Moreover, channels that hold photopatterning capability are highly desired to minimize parasitic capacitance and simplify the fabrication process/cost. However, yielding photopatternable channels with both precise/robust patterning capability and controllable ionic-electronic coupling is still challenging. Herein, double-end trifluoromethyl diazirines (DtFDA) with different chain lengths are introduced in the OECT channel to act as both photocrosslinker and medium to regulate ionic-electronic transport. Specifically, high-resolution patterns with a minimum line width/gap of 2 μm are realized in p(g2T-T) or Homo-gDPP based channels by introducing DtFDA. Maximum transconductances of 68.6 mS and 81.6 mS, current on/off ratio of 10^6 and 10^7 (under a drain voltage of only ±0.1 V), are achieved in p- and n-type vertical OECTs (vOECTs), respectively, along with current densities exceeding 1 kA cm^{-2} and good cycling stability of more than 100,000 cycles (2000 seconds). This work provides a new and facile strategy for the fabrication of vOECT channels with high resolution and high performance via the introduction of a simple and efficient crosslinker.

Introduction

In 1947, Bell Labs reported the first transistor, ushering in the era of microelectronics and revolutionizing our lives.[1] Over the past 70 years, various inorganic and organic semiconductors have been developed. In contrast to inorganic semiconductors, organic semiconductors hold diverse advantages, including solution processability, lightweight, mechanical flexibility, and low-cost fabrication over large areas.[2–5] As a representative of organic transistors, organic electrochemical transistors (OECTs), which enable bulk ionic doping/de-doping of the transistor channel,[6–8] were developed by Wrighton in 1984.[9] After decades of development on both advanced organic mixed ionic-electronic (semi)conductors (OMIECs) and novel device structures, excellent characteristics in OECTs, including low drive voltage (<1 V), high transconductance (g_m), stable operation in physiological environments, and good logic circuit performance, have been exhibited. Therefore, they have unique advantages and great potential in the field of extremely sensitive biochemical sensors, neuromorphic devices, and implantable electronics.[10–17]

Differing from organic field-effect transistors (OFETs), the operating principle of OECTs depends on ionic doping/dedoping from the electrolyte to the transistor channel bulk. Therefore, constructing an efficient ionic transport pathway in the channel while maintaining high electron/hole carrier mobility is vital for OECTs.[18–19] For instance, porous semiconducting polymer films with high surface-to-volume ratios and interconnected electron/hole and ion transport paths were introduced not only for high-performance OECTs but also can enhance the flexibility/stretchability of relevant devices.[20] Two new fused bithiophene imide polymers, f-BTI2TEG-T, and f-BTI2TEG-FT, were also synthesized, which can regulate the intermolecular aggregation to construct better ionic transport channels.[21] However, the intrinsic design principle conflict between the highly ordered crystallization structure and the high-porosity structure of conventional semiconducting polymers has led to obstacles in the synergistic improvement of ionic-electronic coupling. In addition, because of the high volume capacitance (C*) of most OMIECs, the OECT channel needs to be precisely patterned and encapsulated to avoid parasitic capacitance. While due to the lack of chemical orthogonality between OMIECs and traditional photoresistors,[22–23] in conventional OECT fabrication process, complicated photolithography process along with multiple etching or stripping processes are required to yield fine patterning of the OMIEC channel.[24] Note, photo-crosslinkers have been utilized to alter chain packing and aggregation of conjugated organic molecules in OFET,[25] but their effectiveness in OMIECs-based OECT channels remains unclear.

[*] Y. Lai, J. Cheng, G. Zhu, Prof. L.-W. Feng
Key Laboratory of Green Chemistry & Technology, Ministry of Education, College of Chemistry, Sichuan University, Chengdu, 610065, China
E-mail: liangwenfeng@scu.edu.cn
M. Xie, Prof. W. Huang
School of Automation Engineering, University of Electronic Science and Technology of China (UESTC), Chengdu, 611731, China
E-mail: whuang@uestc.edu.cn
Prof. J. Chen
Department of Chemical Science and Technology, Yunnan University, Kunming, China

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In recent years, several crosslinking agents have been developed for OECTs.[26–27] Especially, OMIECs blended with photo-crosslinkers through direct photo patterning in OECTs have been reported.[28–29] Nevertheless, these studies primarily utilized photo-crosslinkers for channel patterning purposes, without a detailed examination of how crosslinkers impact morphology, aggregation, and ionic/electronic properties.

With the rapid development of various OMIEC materials, novel architectures of OECTs are being constantly explored and optimized. In 2018, a vertical OECT architecture was proposed, which exhibits a reduced footprint (W = 70 μm, L = 450 nm–1.4 μm, d = 70 nm–180 nm) along with high g_m by vertically stacking the source and drain electrodes with an insulating Parylene-C layer in between.[30] After that, various vertical architectures have been developed and shown tremendous advantages for various applications.[31–32] In 2023, a newly invented vertical OECTs architecture (vOECTs, where the transistor channel is directly sandwiched in between vertically stacked source/drain electrodes) shows ultra-high g_m (0.2–0.4 S), and cycling stability (> 50000 cycles) for both p- and n-type devices.[33] In such architecture, a direct photo-patternable channel was realized by introducing cinnamate-cellulose (Cin-Cell) (Figure 1a) as the crosslinker,[34] which also extensively simplified the fabrication process of the channel. However, the synthesis process of Cin-Cell is relatively complicated and the capability of such crosslinker to yield precise patterns without the aid of a photo initiator is still unclear. Moreover, Cin-Cell as a photo crosslinker suffers from a narrow processing window, which requires a short development time of 1–2 s and a precise blend ratio to avoid negative effects on both ionic and electronic transport. Consequently, photo crosslinkers with a simple synthesis process, robust processing procedure, and precise patterning, are urgently needed.

Herein, double-end trifluoromethyl diazirine (DtFDA) with different intermediate alkyl chain lengths are synthesized and utilized as the photo crosslinker of OECT channels based on p-type p(g2T-T) or n-type Homo-gDPP (Figures 1b–e). Aggregations, packing orders, and morphologies of OMIEC channels can be effectively modulated by adding DtFDA with varying chain lengths, which are systematically characterized by grazing incidence X-ray diffraction (GIXRD), ultraviolet–visible spectroscopy (UV/Vis), transmission electron microscope (TEM), atomic force microscope (AFM), etc. Therefore, the introduction of DtFDA not only yields micrometer scale patterning of the OMIEC channel but also can regulate ionic-electronic coupling, which leads to controllable ionic doping/dedoping process and electron/hole carrier transport. Specifically, peak g_m ($g_{m,peak}$) and turn-on transient times (τ_{on}) of p-type

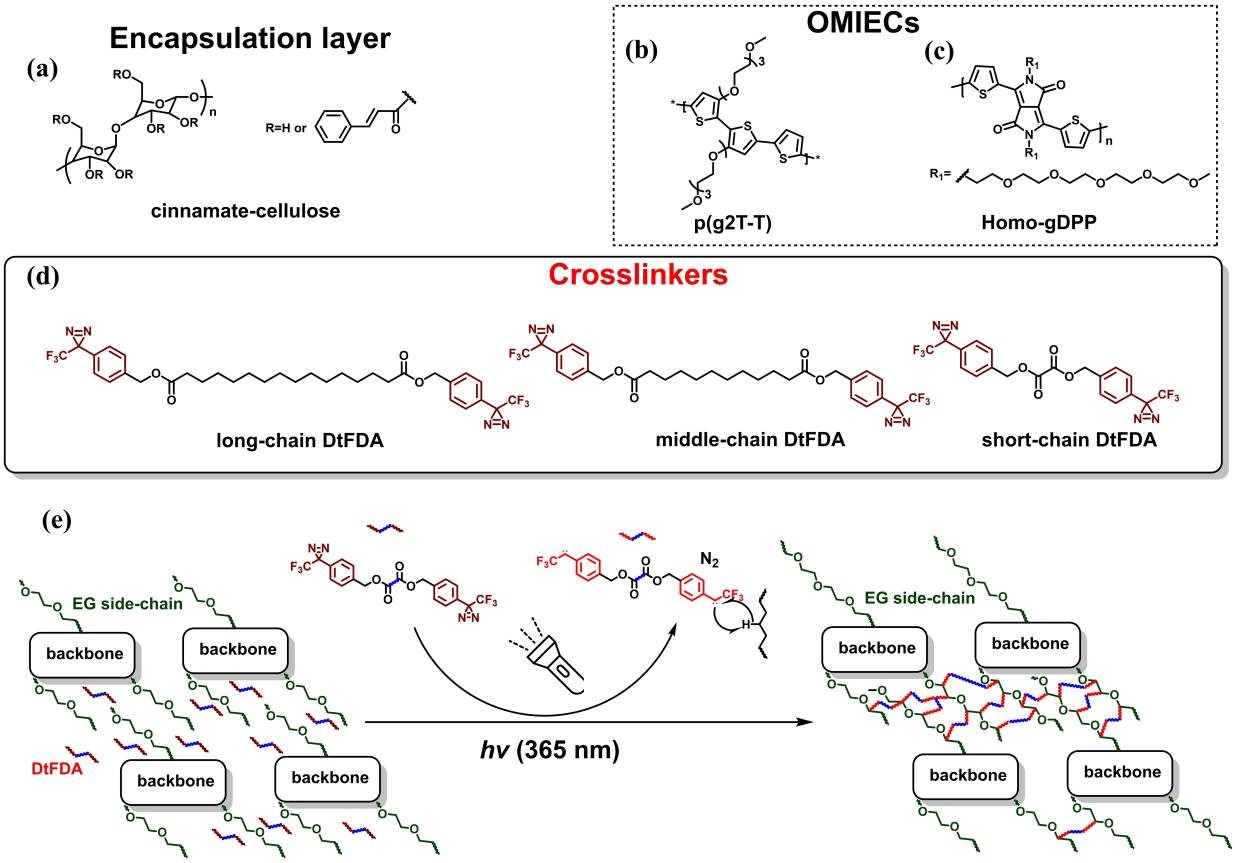


Figure 1. Chemical structures of (a) cinnamate-cellulose, (b) p(g2T-T), (c) Homo-gDPP, and (d) DtFDA with three different chain lengths. (e) Cross-linking mechanism between OMIECs and DtFDA.

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p(g2T-T) based OECTs can be modulated in a wide range of 21.3–68.6 mS, and 0.93–9.56 ms, respectively, by incorporating DtFDA with different chain lengths. Noteworthy, the OECTs based on p(g2T-T) with long-chain crosslinkers have a high current switching ratio (I_{ON}/I_{OFF}) of 10^6 and high stability of more than 100,000 cycles (2000 s). For the n-type Homo-gDPP based vOECTs, $g_{m,peak}$ and τ_{on} can be modulated in a wide range of 27.0–81.6 mS and 0.32–0.92 ms, respectively. Interestingly, for the two different OMIECs, performance variation trends along with different chain lengths in DtFDA are different. The packing order of p(g2T-T) becomes stronger when mixing with longer chain crosslinkers, while in Homo-gDPP, mixing with longer chain crosslinkers results in weaker packing order. In general, stronger packing is conducive to electron/hole carrier transport, which is manifested by higher I_{on} and g_m. Therefore, the performance of both p- and n-type OECTs can be effectively modulated by changing the length of the crosslinker intermediate chain, along with high-precision patterning of the channel. This is the first report on the regulation of channel material stacking and ionic-electronic coupling through photo crosslinkers in OECTs. Moreover, the DtFDAs can be synthesized in short steps (one or two) with high yield (up to 94 %), and all the starting materials are commercially available. This work not only introduces an effective way for precise patterning of microelectronics but also proposes a facile tool to optimize the micro/nano structures and ionic/electronic properties of organic functional materials.

Results and Discussion

Diazirine has been used in the field of biological protein labeling for several years,[35] while due to its high reactivity, diazirine analogue has been gradually developed as a crosslinker for aliphatic polymers[36] and other functional materials.[37] Diazirine is deemed "electronically clean" as the light-triggered crosslinking will react with the non-conjugated side chains, without breaking the conjugation in the backbone.[38,39] It has been applied for the direct lithography of active layers in OFETs[38–39] and organic light-emitting diodes (OLEDs).[37] Here, DtFDA is designed with three different intermediate chain lengths (Figure 1d) and synthesized by a one-step substitution reaction. Such crosslinkers hold three major advantages: 1) Photoreactive diazirine group has high reactivity, and can be directly crosslinked with semiconductor;[38] 2) Raw materials of the crosslinker are commercially available, and few synthesis steps are needed; 3) There are no extra functional groups in the crosslinkers except trifluoromethyl as a stable group, which ensure the subsequent changes of channel morphology and ionic/electronic properties be influenced only by chain length regulation. The detailed synthesis routes are provided in the Supporting Information (Scheme S1). Note, the synthesis of crosslinkers is gram-scale and convenient, along with a high yield of >90 %.

The heat stability of these DtFDA crosslinkers is first investigated by thermogravimetric analysis (TGA) (Figure S1). Decomposition temperatures of the diazirine groups in all three crosslinkers are above 100 °C, which ensures unnecessary thermal crosslinking will not occur during the fabrication process of OECTs (especially during the preparation and storage of the blend solutions). Absorption spectra of the DtFDA are next analyzed by UV/Vis, and they determine the required wavelength for crosslinking and also enable qualitative analysis of their aggregations (Figures 2a–b). The maximum absorption ranges of all the crosslinkers in both solution and film are between 330 nm and 370 nm, which is attributed to the n-π^* transition of diazirine. When the diazirine is exposed to UV light (here 365 nm UV lamp with 600–800 mW/cm^2 is used), the diazirine photo-activable group in the crosslinker releases nitrogen and forms carbene. Meanwhile, carbene is then quickly inserted into the C–H bond of the polymer to achieve crosslinking.[39] Note, Cin-Cell crosslinks under 280 nm UV light, whereas the longer UV wavelength (365 nm) utilized for DtFDA will minimize the damage to the semiconducting polymer by introducing lower photon energy.[40–41]

To further determine the crosslinking efficiency, a UV/Vis spectrometer is further utilized to characterize the absorption intensity of patterned p(g2T-T):DtFDA blends (The blend films are photo-crosslinked and further developed; Detailed fabrication process can be found in SI). Note, since the absorption intensity of compounds can be used as the basis for substance quantification (the more OMIEC remains after development, the higher the absorbance retention should be observed), the crosslinking/patterning efficiency is then evaluated by the degree of reduction on the maximum absorption peak intensity of p(g2T-T) at ~602 nm. Different blend ratios and UV exposure time of the p(g2T-T):DtFDA blends are screened (Figures 2c–d, S2, S3, and S4), where crosslinking/patterning efficiency increases with the crosslinker mass ratio and UV exposure time. Especially, 3 mins of UV exposure and p(g2T-T):DtFDA blend ratio of 1:1 (in mass) leads to more than 90 % retention of the original film (Figure 2d). While Homo-gDPP retained slightly less mass (78 %) than p(g2T-T) after crosslinking and development, due to better solubility of Homo-gDPP in chloroform. Note, the short-chain DtFDA requires less exposure dose to achieve effective crosslinking (Figure S5), due to the higher mass ratio of photoreactive groups in the molecule. While the maximum absorptions of the two blends remain identical after exposure (Figure S6), which indicates that DtFDA crosslinks with alkoxy side chains rather than the conjugated backbone (Figure 1e). Since DtFDA crosslinks with semiconductors under UV light exhibit good resistance to organic solvents, we further studied the patterning resolution of such system. Various patterns based on OMIEC:DtFDA blends, including irregular shapes and text letters, are precisely patterned with a minimum line width/gap of 2 μm (Figures 2e–f, S7, and S8). Except for p(g2T-T) and Homo-gDPP, other two semiconducting polymers (N2200 and P3HT) can also form accurate patterns with high resolution by adopting DtFDA crosslinker (Figures S9–S11), which proves that DtFDA shows good universality for

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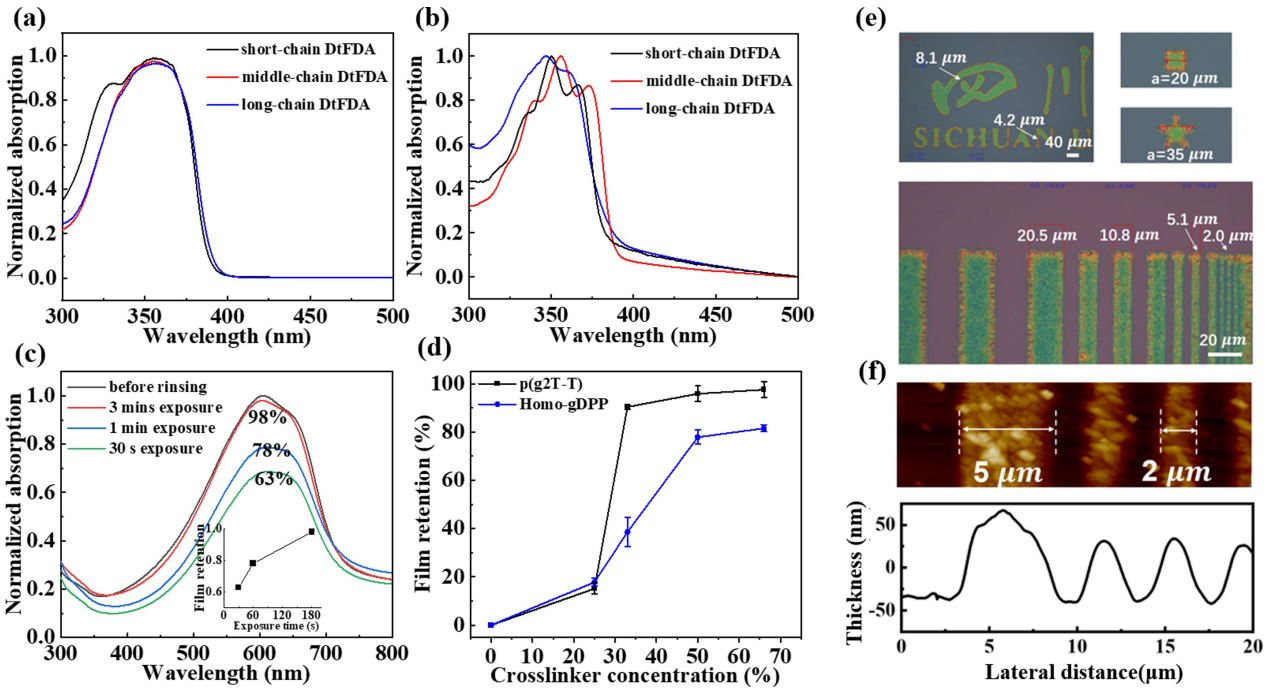


Figure 2. Absorption spectra of DtFDA in (a) chloroform and (b) film state. (c) Absorption spectra of p(g2T-T): long-chain DtFDA blend films (1 : 1 in mass) with different UV exposure times. (d) Film retention of p(g2T-T): long-chain DtFDA blend [0, 20, 33, 50, and 66 wt% of DtFDA in the p(g2T-T) : DtFDA blend] after 3 mins of UV exposure and $CHCl_3$ developing. Note that, film retention percentage is the ratio of crosslinked film maximum absorption wavelength before developing versus after developing. (e) Optical microscope images and (f) AFM height image of patterned p(g2T-T): long-chain DtFDA blend films (1 : 1 in mass).

semiconducting polymer patterning and has the potential for the fabrication of various devices that require accurate patterning.

GIXRD is next used to determine the solid-state packing structure of different OMIEC films. As shown in Figure 3a, pristine p(g2T-T) film shows lamellar packing in out-of-plane direction, where $q_z = 0.45$ Å^{-1} (100), 0.92 Å^{-1} (200) (*d*-spacing = 13.6 Å, 6.82 Å, respectively). While no obvious diffraction peaks in in-plane direction are observed, which indicates an "edge-on" dominated orientation in pristine p(g2T-T). After blended with DtFDA [p(g2T-T) : DtFDA = 1 : 1 in mass] and UV-crosslinked, diffraction intensities and preferred orientation of the films are weakened. Especially, for all the DtFDA with different chain lengths, (200) and (300) diffraction peaks ($q_z = 0.92$ Å^{-1}, 1.39 Å^{-1}) disappear. On the other hand, (100) diffraction peak ($q_z = 0.45$ Å^{-1}) shows gradually weakened intensity with shorter DtFDA intermediate chain length. Therefore, the introduction of DtFDA makes the stacking of p(g2T-T) more disordered, especially with DtFDA holds a shorter intermediate chain length. It is speculated that a shorter intermediate chain length of DtFDA allows less degree of freedom during the blending and cross-linking process. While when blending with long-chain DtFDA, p(g2T-T) holds more freedom to move around, leading to easier stacking in the preferred dominant direction Similarly, pristine Homo-gDPP film shows a characteristic scattering pattern in agreement with a preferential edge-on molecular packing (Figure 3b), where strong out-of-plane scattering peaks ascribed to the lamellar repetition (100 and 200 peaks) with spacing of 20.4 Å and 10.3 Å ($q_z = 0.31$ Å^{-1} and 0.61 Å^{-1}) are observed. After blended with DtFDA and UV-crosslinked, gradually weakened crystallinity can be observed with increased DtFDA chain lengths. Furthermore, with the increasing of chain length in DtFDA, increased *d*-spacings are observed simultaneously, where lamellar spacings of the (100), (200) increase from 20.3 Å and 10.3 Å (pristine Homo-gDPP) to 21.7 Å and 11.0 Å (with short-chain DtFDA), 22.8 Å and 11.8 Å (with middle-chain DtFDA), 23.2 Å and 11.6 Å (with long-chain DtFDA), respectively (Figure 3d). Such an interesting trend in *d*-spacing variation could be due to the fact that, compared to long-chain DtFDA, short-chain DtFDA could lead to tighter packing of Homo-gDPP. Moreover, with short-chain DtFDA, the direction of lattice stacking also changed to a certain extent. Specifically, the lamellar packing of Homo-gDPP in the in-plane direction (100) disappears, which indicates that Homo-gDPP is more ordered in the out-of-plane direction when blending with short-chain DtFDA. Therefore, for p(g2T-T), by reducing the chain length of DtFDA, reduced crystallinity in the blend film is observed. While for Homo-gDPP, reduced crystallinity is obtained in all blend films, along with regulated lattice spacing distance and slight change to face-on orientation when adopting DtFDA with different chain lengths.

AFM is further utilized to study the surface morphology, roughness, and phase separation of the crosslinked blend films. For p(g2T-T), the one blended with short-chain

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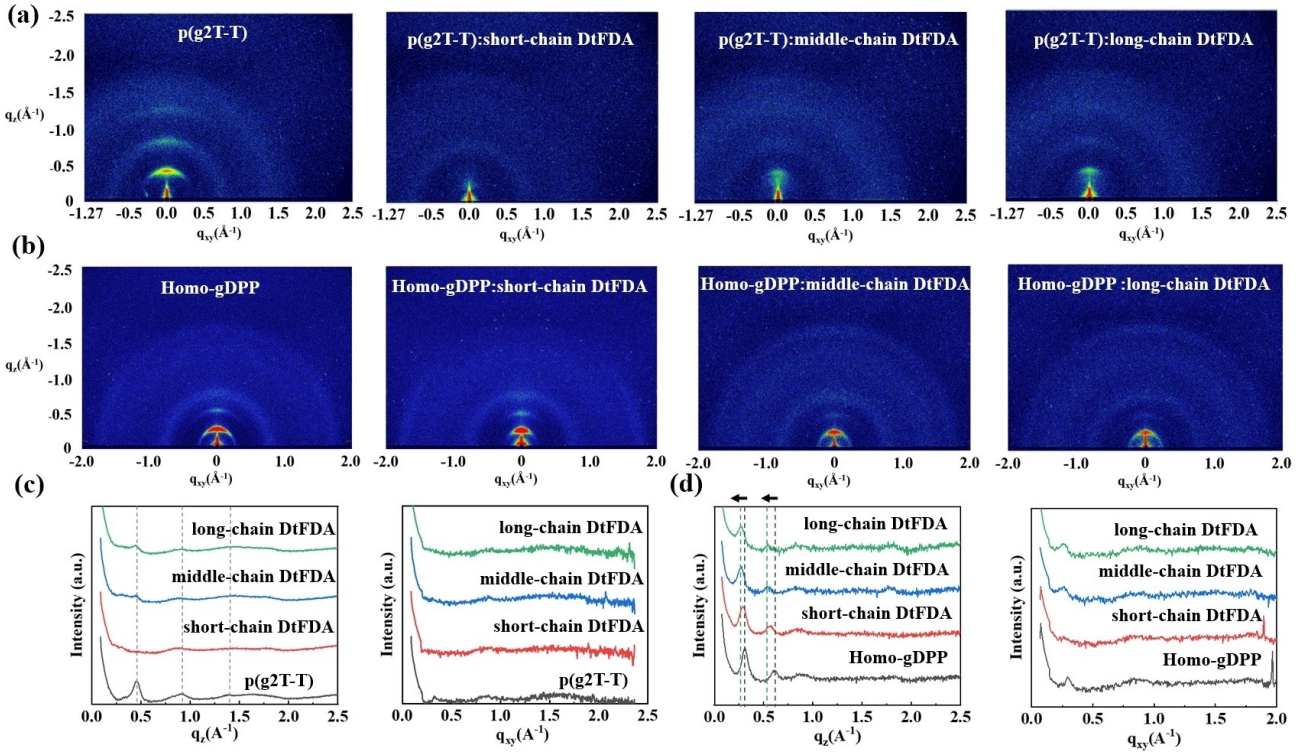


Figure 3. (a, b) 2D GIXRD patterns and (c, d) linecuts for OMIECs : DtFDA blend films with different DtFDA intermediate chain lengths. (a, c) are for p(g2T-T) : DtFDA blends while (b, d) are for Homo-gDPP : DtFDA blends. Note, OMIEC : DtFDA = 1 : 1 in mass.

DtFDA shows a rougher surface with a root-mean-square roughness (σ_{rms}) of 11.5 nm, while blend films with middle and long-chains of DtFDA show σ_{rms} of 2.68 and 4.08 nm, respectively (Figure S12). For Homo-gDPP (Figure S13), σ_{rms} are 18.3, 9.54, and 6.50 nm for blend films with short, middle, and long chain DtFDA, respectively. Interestingly, no obvious phase separation between OMIEC and different DtFDA is observed, making the blend be solid solution. Although varying the chain length of DtFDA can lead to a slight change in the morphology of the blend film, such a change in surface morphology is not essential. TEM is next applied to characterize the crosslinked blend films. Similar to the results in AFM, a uniform mixture is observed in all p(g2T-T) : DtFDA blends(Figure S14). Moreover, the lattice domain size decreases gradually as longer chain DtFDA is added. The smaller lattice domain of the blend with long-chain DtFDA indicates that it has a more evenly distribution of p(g2T-T) and DtFDA, which may allow ions to permeate rapidly in the bulk membrane thus enabling fast transient speed and good electrochemical doping efficiency. All Homo-gDPP : DtFDA blends also show a uniform mixture, where no phase separation and no clear domain size variation are observed (Figure S15). Finally, water contact angle measurements (Figure S16) indicate that all of the crosslinked p(g2T-T) films are far more hydrophobic ($\approx 90°$) than pure p(g2T-T) ($\approx 70°$). It is possible to reduce the excessive water taken up by the film which would limit the speed of electrochemical switching and impair the reversibility of the doping process.[42]

Based on the above characterizations, it is concluded that DtFDA can achieve accurate patterns when mixing with different OMIECs, and lead to unique microstructure manipulation. Therefore, OMIEC based channel and Cin-Cell based encapsulation layer with DtFDA as crosslinker are introduced for the fabrication of vOECTs (Figure 4a, detailed fabrication process and device optical microscopic images can be found in Figures S17 and S18). Briefly, for vOECTs, chromium (Cr)/gold (Au) are evaporated with a shadow mask and act as a source electrode (40 μm width); next, channel layer based on either p(g2T-T) : DtFDA blend or Homo-gDPP : DtFDA blend is spin-coated and photo-patterned (200×200 μm), followed by the evaporation of a top Au drain electrode (40 μm width); then, the encapsulation layer (Cin-Cell : DtFDA = 4 : 1 in mass) is spin-coated and photopatterned, where a 200×200 μm square is left open on top the of channel layer to provide passways for electorlyte injection/ejection. By introducing Cin-Cell as the encapsulation layer covering gold electrodes, formation of the electrical double layer around the electrode can be effectively eliminated (Figure S19), thereby avoiding the formation of addition parasitic capacitances between the electrolyte and electrode, decreasing the gate current, and improving the stability of the device. Lastly, an electrolyte solution (0.01 M phosphate buffer solution (PBS)) is added to the encapsulation layer opening area along with the insertion of Ag/AgCl as the gate electrode.

According to previous bandwidth and electrochemical impedance spectroscopy (EIS) measurements,[33] vOECT,

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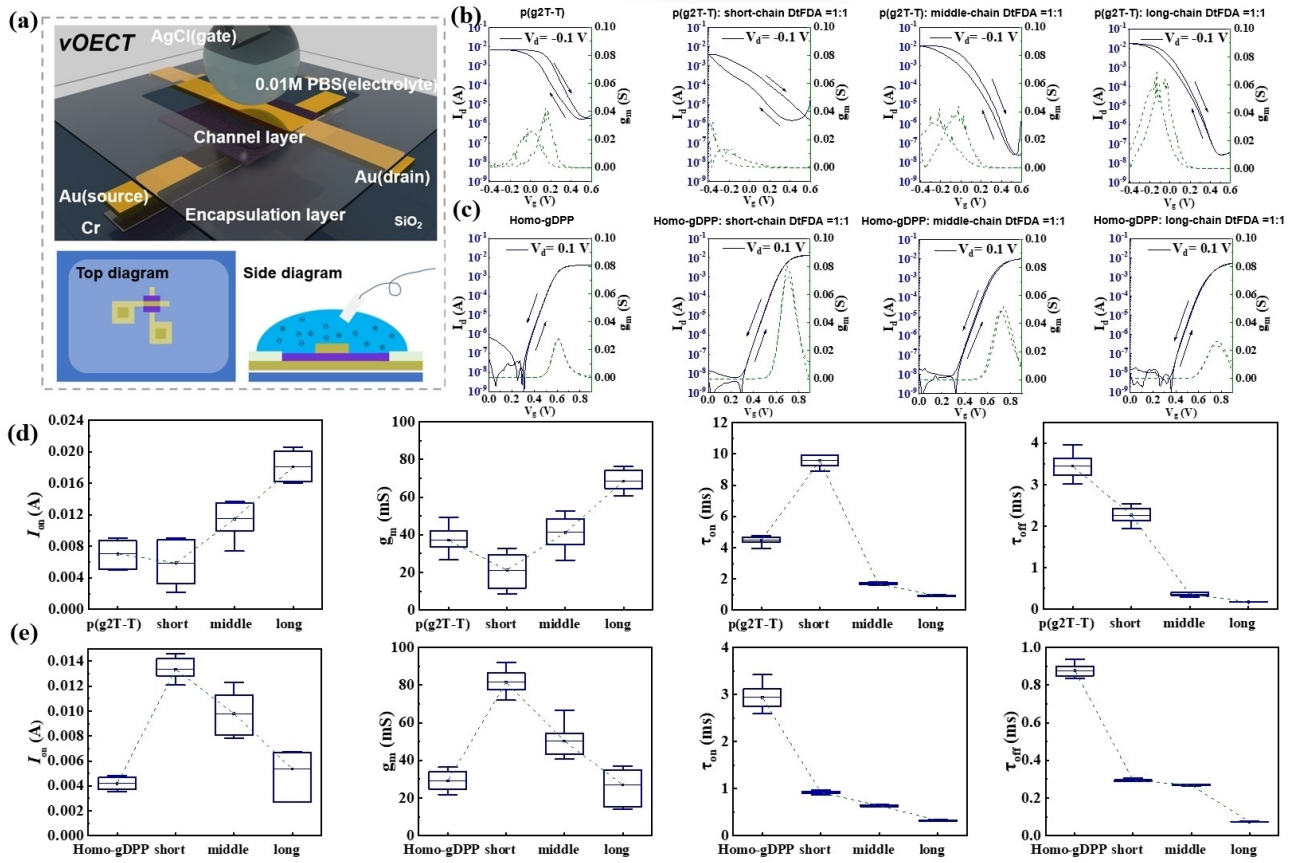


Figure 4. vOECT structure and transistor performance characteristics. (a) Disassembly diagram, top-view, and side-view diagrams of vOECT. Transfer and transconductance characteristics of pure or patterned p(g2T-T) (b) and Homo-gDPP (c). Dependencies of I_{on}, $g_{m,}$ $\tau_{on,}$ and τ_{off} in p-type (d) and n-type (e) vOECTs on the different chain lengths of DtFDA.

compared to conventional planar OECT, not only reduces the length of channels and footprints, but also induces a strong electric field which could enhance ionic drift velocity. Consequently, in the current study, we fabricated a vOECT structure to verify the properties of the OMIEC:DtFDA blends.

vOECTs based on p(g2T-T):DtFDA blends, where three DtFDA with different chain lengths are blended with p(g2T-T) in a mass ratio of 1:1, respectively, are characterized. For each kind of vOECTs, at least 8 devices are measured and analyzed, and corresponding transistor parameters are summarized in Table 1 and Figures 4, S20, and S21. Note, the transfer characteristics are collected on the 3rd scan to ensure stable characteristics. Peak g_m are collected from the forward sweeps of transfer curves. To quantify the hysteresis, the voltage differences are obtained by calculating the halfway current between the on and off states ($|I_{DS}| = 10^{-5}$ A for p-type and n-type).[43–44]

Utilization of DtFDA as a crosslinker along with the patterned channel leads to tremendously different transfer characteristics. First, I_{on} of vOECT based on pure p(g2T-T) is 7.04 ± 1.77 mA (Note, for such transistor the channel is not patterned due to lack of photo-crosslinking capability. Therefore, only a rough comparison can be conducted with other patterned devices) After patterning of the OMIEC and applying encapsulation, effectively enhanced I_{on}/I_{off} ratio is achieved along with decreased I_{off}. Moreover, gate currents of the vOECTs also decrease with patterned OMIECs and encapsulation (Figure S22), suggesting lower parasitic capacitance after introducing patterned OMIECs and encapsulation layer. While vOECTs with p(g2T-T): short-chain DtFDA show similar I_{on} of 5.89 ± 2.99 mA. Interestingly, with the increasing of chain length of DtFDA, enlarged I_{on} of 11.5 ± 2.3 mA (with middle chain DtFDA) and 18.1 ± 2.9 mA (with long-chain DtFDA) are obtained. Meanwhile, I_{off} of vOECT with pure p(g2T-T) is 10^{-6} A, which is similar to that in vOECT with short-chain DtFDA, along with obvious hysteresis. Switching to p(g2T-T):DtFDA with middle or long chain, decreased I_{off} (10^{-8} A) is observed, along with minimized hysteresis. As shown in Figure 3c, p(g2T-T) lattice packing and lattice domains change regularly with the change of the DtFDA chain length, which correlates with the performance variation of the vOECT based on DtFDA with different chain lengths. Specifically, a similar tendency in peak g_m is also observed, where it increases significantly from 21.3 ± 9.8 mS (with short-chain DtFDA) to 41.3 ± 9.3 mS (with middle-chain DtFDA), and 68.6 ± 5.8 mS (with long-chain DtFDA). Such results indicate that ions from the electrolyte can inject or eject more quickly and efficiently in the channel with

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Table 1: Summary of transistor parameters of both p- and n-type vOECTs based on OMIEC : DtFDA and OMIEC:Cin-Cell blends.

Material	I_{on} [A]	I_{off} [A]	$g_{m,peak}$ [mS]	τ_{on} [ms]	τ_{off} [ms]	V_{on} [V]	Hysteresis [mV]
p(g2T-T)[a]	$(7.04\pm1.77)\times10^{-3}$	$(1.67\pm0.67)\times10^{-6}$	37.1 ± 7.2	4.47 ± 0.27	3.44 ± 0.30	0.50 ± 0.01	80.3 ± 18.7
p(g2T-T): short chain	$(5.89\pm2.99)\times10^{-3}$	$(1.36\pm2.48)\times10^{-6}$	21.3 ± 9.8	9.56 ± 0.38	2.26 ± 0.20	0.45 ± 0.06	240.0 ± 61.4
p(g2T-T): middle-chain	$(1.15\pm0.23)\times10^{-2}$	$(2.47\pm0.67)\times10^{-8}$	41.3 ± 9.3	1.70 ± 0.07	0.36 ± 0.04	0.54 ± 0.02	56.9 ± 29.6
p(g2T-T): long-chain	$(1.81\pm0.29)\times10^{-2}$	$(2.70\pm1.03)\times10^{-8}$	68.6 ± 5.8	0.93 ± 0.03	0.18 ± 0.01	0.50 ± 0.03	23.8 ± 16.9
p(g2T-T): Cin-Cell[b]	$(1.26\pm3.20)\times10^{-2}$	$(3.04\pm3.09)\times10^{-8}$	51.9 ± 13.5	5.30 ± 0.32	2.58 ± 0.13	0.52 ± 0.01	109.3 ± 44.5
Homo-gDPP[a]	$(4.15\pm0.47)\times10^{-3}$	$(3.43\pm3.15)\times10^{-8}$	29.1 ± 5.4	2.94 ± 0.27	0.88 ± 0.03	0.29 ± 0.02	2.8 ± 0.7
Homo-gDPP: short-chain	$(1.34\pm0.77)\times10^{-2}$	$(3.46\pm2.70)\times10^{-9}$	81.6 ± 6.8	0.92 ± 0.03	0.29 ± 0.01	0.29 ± 0.03	3.4 ± 2.2
Homo-gDPP: middle-chain	$(9.67\pm1.55)\times10^{-3}$	$(7.09\pm5.68)\times10^{-9}$	50.3 ± 8.6	0.64 ± 0.02	0.27 ± 0.01	0.31 ± 0.03	9.4 ± 4.1
Homo-gDPP: long-chain	$(5.07\pm1.88)\times10^{-3}$	$(5.18\pm1.71)\times10^{-9}$	27.0 ± 10.1	0.32 ± 0.01	0.07 ± 0.01	0.36 ± 0.03	5.6 ± 2.2

[a] The channels of vOECTs with pristine OMIECs are not patterned. [b] p(g2T-T):Cin-Cell = 9 : 2 in mass. All parameters are based on more than 8 devices.

longer chain DtFDA, leading to a faster and more complete doping/dedoping process of the channel. This is further proved by τ_{on} and turn-off transient time ($\tau_{off,}$) of the vOECTs, where they decrease from 9.56 ± 0.38 and 2.26 ± 0.20 ms (with short-chain DtFDA), to 1.70 ± 0.07 and 0.36 ± 0.04 ms (with middle-chain DtFDA), and 0.93 ± 0.03 and 0.18 ± 0.01 ms (with long-chain DtFDA), respectively (Figures 4d, S23 and Table 1). As a result, tremendously enhanced device performance is obtained in vOECT based on p(g2T-T) : DtFDA channel where long-chain DtFDA is applied. Although vOECT based on Cin-Cell shows little difference to long-chain DtFDA in g_m and current on-off ratio under optimal conditions (Figures 4d, S24), it exhibits significantly greater hysteresis (109.3 mV) and response time ($\tau_{on}=$ 5.30 ms, $\tau_{off}=2.58$ ms). It could be due to the special cycloaddition reaction mechanism of self-crosslinking, resulting in an obvious phase separation when mixed and cross-linked with semiconductors. Moreover, for Cin-Cell as crosslinker, with an optimal blend ratio of 9 : 2,[33] less than 50 % of the original film is retained after developing for 5 s (Figure S25). This severely limits its processing window (only 1–2 s), while the ones with DtFDA as the crosslinker, show robust interconnected crosslinking up to 60 mins (Figure S26). In addition, an increase in Cin-Cell content will further increase the cross-linking efficiency, but also will destroy effective ionic and electronic transport paths, which will deteriorate the OECT performance obviously (Figure S24).

Next, OECTs based on n-type Homo-gDPP are fabricated and characterized through the same process. Similarly, the utilization of different chains of DtFDA as crosslinkers along with the patterned channel leads to tremendously different transfer characteristics. I_{on} of vOECT based on pure Homo-gDPP is 4.15 ± 0.47 mA. While vOECT with Homo-gDPP:long-chain DtFDA shows a similar I_{on} of 5.07 ± 1.88 mA. While decreasing DtFDA chain length leads to an enlarged I_{on} of 9.67 ± 1.55 mA (with middle-chain DtFDA) and 13.4 ± 0.77 mA (with short-chain DtFDA). As shown in Figure 3d, Homo-gDPP lattice packing strengthens regularly with the decreasing chain length of the crosslinker, which correlates well with the performance variation of the vOECT. A similar tendency in peak g_m is also observed, where it increases significantly from 27.0 ± 10.1 mS (with long-chain DtFDA) to 50.3 ± 8.6 mS (with middle-chain DtFDA), and 81.6 ± 6.8 mS (with short-chain DtFDA). Similar to p(g2T-T), τ_{on} and τ_{off} become shorter with increasing DtFDA chain length, where τ_{on} and τ_{off} decrease from 0.92 ± 0.03 and 0.29 ± 0.01 ms (short-chain DtFDA), to 0.64 ± 0.02 and 0.27 ± 0.01 ms (middle-chain DtFDA), to 0.32 ± 0.01 and 0.07 ± 0.01 ms (long-chain DtFDA), respectively (Figures 4e, S27 and Table 1). Based on the above results, it is obvious that tighter lattice packing can enhance the electron transport in the channel, thereby improving I_{ON} and g_m. Less amorphous areas and smaller d-spacing in Homo-gDPP will hinder the penetration of ions in the channel layer to a certain extent, leading to a longer switching time. Besides, with the increasing DtFDA chain length, enlarged V_{on} are observed simultaneously, from 0.29 ± 0.03 V (short-chain DtFDA) to 0.31 ± 0.03 V (middle-chain DtFDA), and 0.36 ± 0.03 mV (long-chain DtFDA). Such a trend could be attributed to the introduction of more alkyl chains.

It is worth noting that, vOECTs based on DtFDA crosslinkers also represent high stability (Figures S28–S29), with no significant current decline after 100,000 cycles (cycle time for 2000 s) (Figure S30). To sum up, the device performance of vOECT can be modulated by adjusting the chain length of the DtFDA crosslinker. This is a new strategy to further improve the performance of OECT. Previously, most researchers focused on the structural modification of semiconductor materials,[45–46] whereas our study reveals that a small change in the structure of the crosslinker can also significantly affect the performance of the device.

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Conclusion

In summary, the synthesis and characterization of a series of diazirine-phenyl-trifluoromethyl-alkyl crosslinkers, along with their utilization in high-performance vOECTs, are reported. Synthesis of these crosslinkers is simple and requires only one step from commercially available raw materials. By changing the length of the intermediate chain, vOECT channels with cooperatively ordered regulation on their ionic/electronic coupling and high patterning precision are realized. Specifically, such small molecule diazirine crosslinkers have high reactivity, high patterning resolution, and no need for additional 'photo-initiators'. It only reacts with the side chains of organic semiconductors, meaning no side reactions and no damage to the conjugated backbone of OMIECs. Consequently, high-performance vOECT has been successfully fabricated showing high g_m (>80 mS), current on/off ratio (10^7), and cycling stability ($>100{,}000$ cycles, and >2000 s). We believe this contribution provides the community with a valuable strategy for developing efficient OECT fabrication processes while introducing a comprehensive approach to constructing organic functional layers with decent UV cross-linking capability, high-resolution patterns, and good stability for bioelectronics.

Supporting Information

The authors have cited additional references within the Supporting Information.[33,47–48]

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available in the supplementary material of this article.

Keywords: organic electrochemical transistor · patterning · chain length · photo-crosslinker · ionic-electronic coupling

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